

SUMMER FELLOWSHIP PROGRAMME
Final Report

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1) Passive RC low pass filter (netlist)

```
r1 1 2 1k
*resistor
c1 2 0 1u
*capacitor
vin 1 0 SIN(0 5v 1khz)
.tran 0.02ms 20ms

*runtime control functions
.control
run
plot v(1) v(2)
.endc
*end control

.end
```

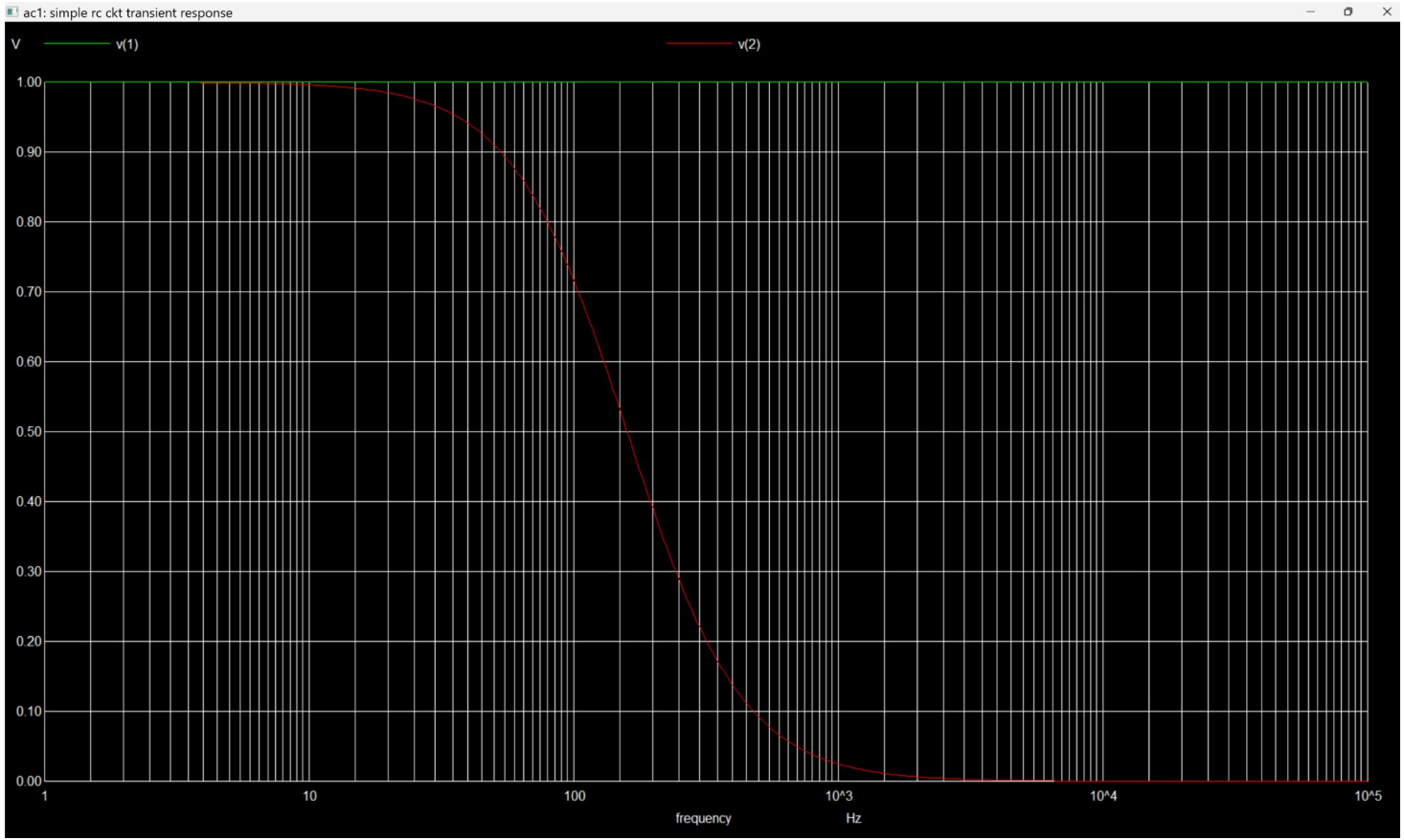
Calculations

R = 1K ohm and C= 1uF

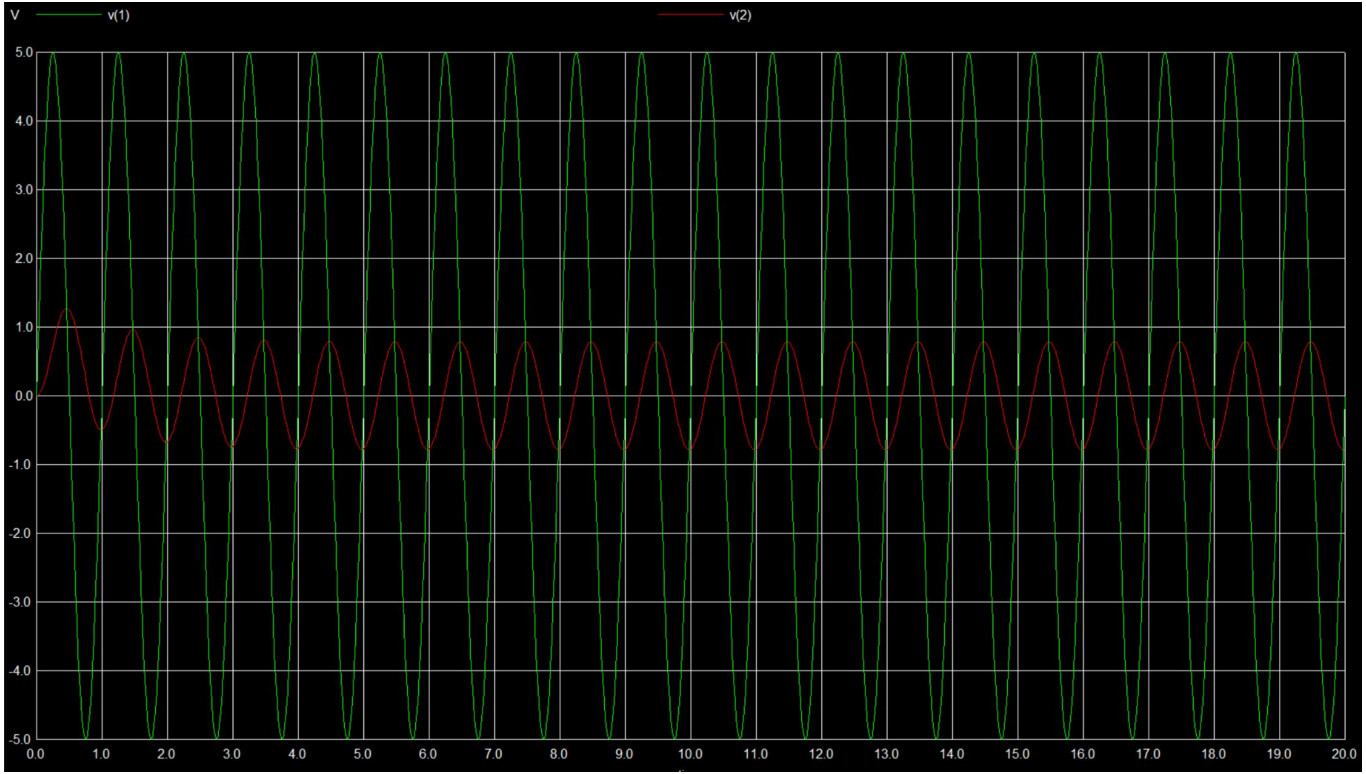
1) Cutoff frequency= 1/(2*pi*rc) = 159.15 Hz
2) Phase delay=-tan^-1(wrc) = -80.9 degrees
3) Time constant RC= 10^-3 seconds

Results match with ngspice simulation results

Output waveforms (frequency response)



Transient response



2) MOSFET (Id vs Vgs) characteristics (netlist)

```
* NMOS DC sweep: Id vs Vgs

.include tsmc018.lib

VDD drain 0 0

VGS gate 0 0

M1 drain gate 0 0 CMOSN W=10u L=0.18u

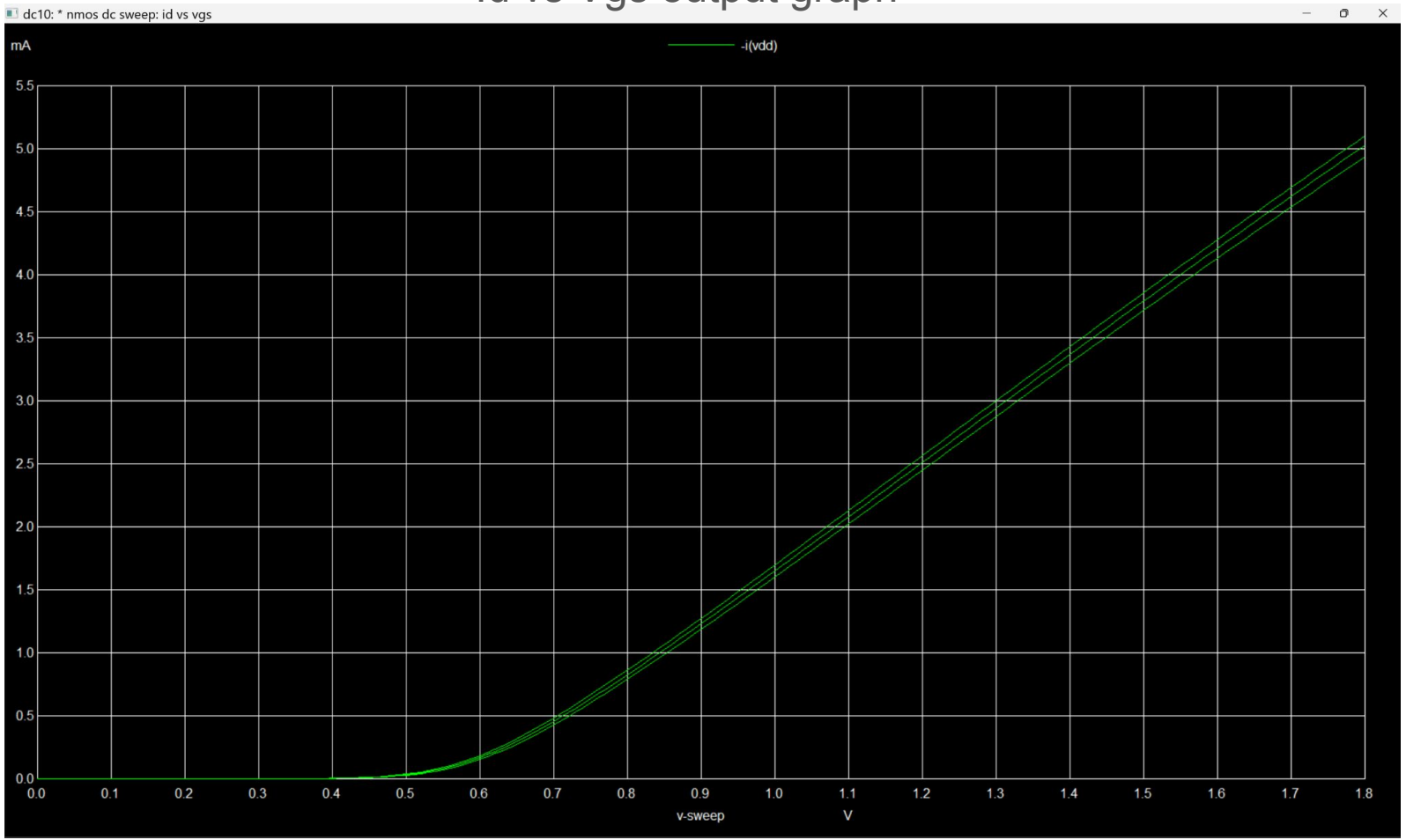
.dc VGS 0 1.8 0.01 VDD 1.2 1.8 0.3
.control
run
plot -i(VDD)
.endc
.end
```

1. VDS = 1.2,1.5,1.8
2. Threshold voltage Vth = 0.366 V
3. Saturation condition:
   VDS ≥ (VGS - Vth)
   At maximum VGS:
   1.8 ≥ (1.8 - 0.366)
   1.8 ≥ 1.434
   Hence transistor operates in saturation region
4. W/L ratio:
   W = 10 μm
   L = 0.18 μm
   W/L = 10/0.18 = 55.56
5. For VGS < 0.366 V:
   Id ≈ 0
6. For VGS > 0.366 V:
   Id ∝ (VGS - 0.366)^2

Hence Id vs VGS graph starts near threshold voltage and rises parabolically.

Results match with ngspice simulation results.

Id vs Vgs output graph



2a) MOSFET (Id vs Vds) characteristics (netlist)

```
* NMOS DC sweep: Id vs Vds

.include tsmc018.lib

VDD drain 0 0

VGS gate 0 0

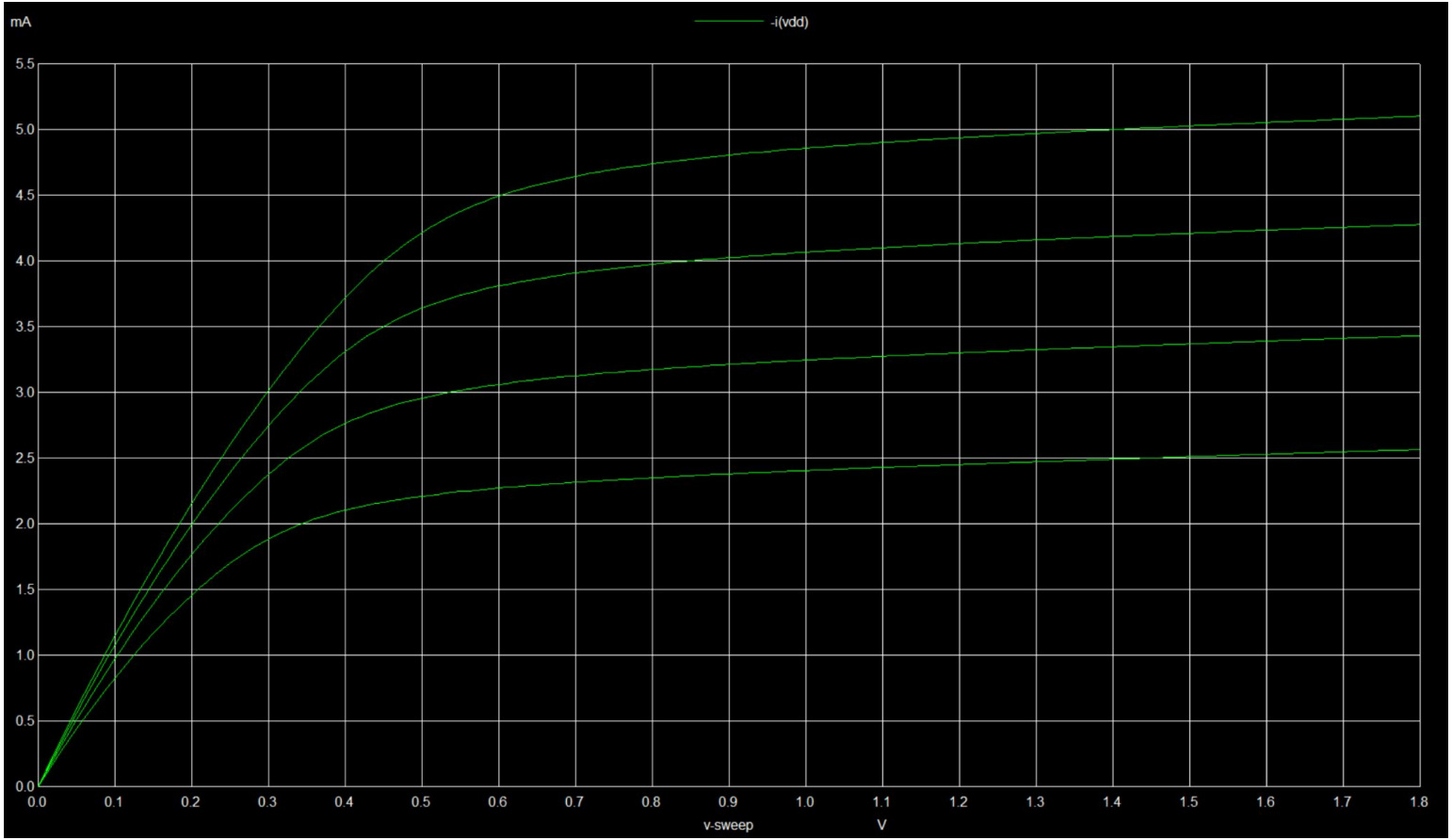
M1 drain gate 0 0 CMOSN W=10u L=0.18u

.dc VDD 0 1.8 0.01 VGS 1.2 1.8 0.3
.control
run
plot -i(VDD)
.endc
.end
```

1. Gate voltage VGS = constant
2. Threshold voltage Vth = 0.366 V
3. Saturation condition:
   VDS >=(VGS - Vth)
4. For VDS < (VGS - Vth):
   MOSFET operates in linear region
5. For VDS <= (VGS - Vth):
   MOSFET operates in saturation region
6. W/L ratio:
   W = 10 µm
   L = 0.18 µm
   W/L = 10/0.18 = 55.56
7. Example for VGS = 1.2 V:
   VDS(sat) = VGS - Vth
   = 0.834 V
   So current increases linearly till 0.834 V and then becomes constant
8. Example for VGS = 1.6 V:
   VDS(sat) = 1.6 - 0.366
   = 1.234 V
   So current increases linearly till 1.234 V and then saturates

Results match with ngspice simulation results.

Id vs Vds output graph



3) Common Source MOSFET Amplifier analysis and Netlist

```
* Common Source Amplifier
.include tsmc018.lib


VDD vdd 0 DC 1.8

VIN in 0 DC 0 AC 1 SIN(0 10m 1k)


CIN in gate 1u


R1 vdd gate 220k
R2 gate 0 100k


RD vdd drain 4.5k

M1 drain gate 0 0 CMOSN W=6u L=0.18u


COUT drain out 1u

RL out 0 100k

.control


op


print v(gate)
print v(drain)
print v(out)
print i(VDD)
print @m1[id]
print @m1[gm]
print @m1[vgs]
print @m1[vds]

tran 1u 5m


plot v(in) v(out)


.endc
```

Calculations

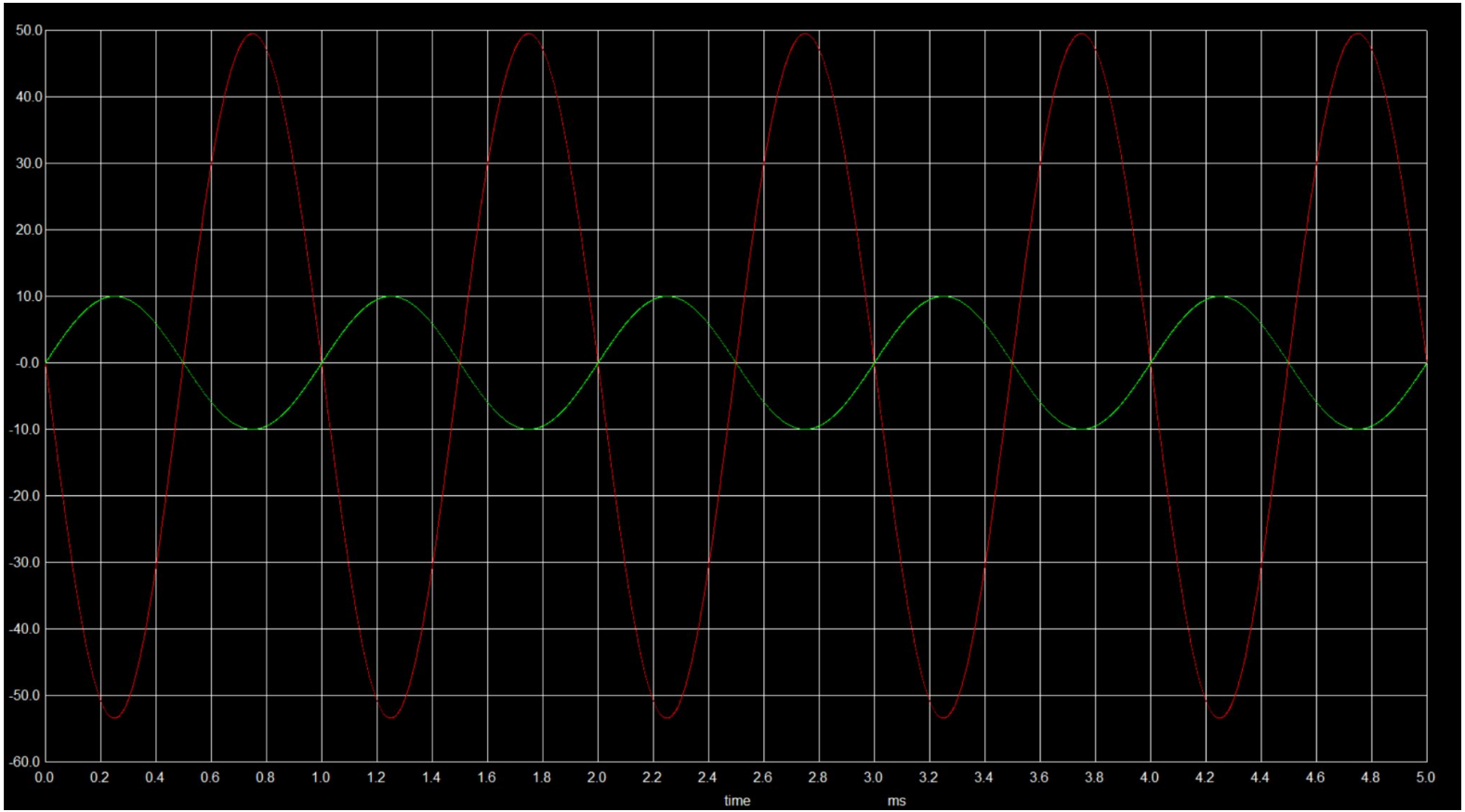
Fixed parameter 1: power consumption (P = VDD × ID)
Fixed parameter 2: target gain=10
Fixed parameter 3:vd~vdd/2=0.9

Solving:

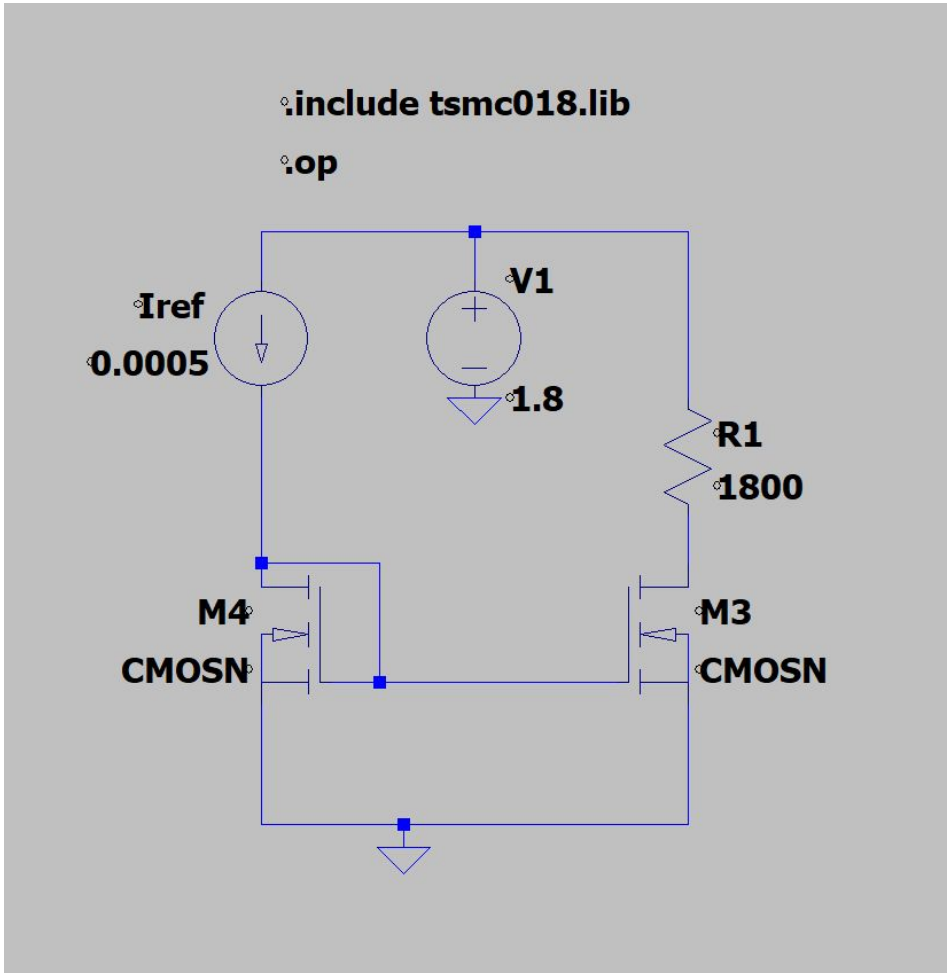
90 µW = 1.8 × ID
ID = 50 µA
VD = VDD / 2
VD = 1.8 / 2
VD = 0.9 V
RD = (VDD - VD) / ID
RD= (1.8 - 0.9) / 50 µA
RD = 0.9 / 50 µA
RD = 18 kΩ
Av = gm × RD
10 = gm × 18k
gm = 10 / 18k
gm = 0.556 mS
gm = 2ID / VOV
0.556 m = 2(50 µA) / VOV
VOV = 100 µA / 0.556 m
VOV = 0.18 V
VGS = VT + VOV

VGS = 0.366 + 0.18
VGS = 0.546 V
gm = μnCox × (W/L) × VOV
0.556 m = 230 μ × (W/L) × 0.18
W/L = 13.43
W = 13.43 × 0.18
W = 2.42 μm
VG = VDD × R2 / (R1 + R2)
0.546 = 1.8 × R2 / (R1 + R2)
R2 / (R1 + R2) = 0.303
R2 = 1.0 megaohm
R1 = 2.3 megaohm
Rin =697 kohm

Input signal and output signal graphs



Minimising power consumption for a current mirror based Cs amplifier (Netlist used)




```
.include tsmc018.lib

.param VDD=1.8
.param ID=0.0152587890625u
.param VD=0.9
.param RDVAL=(VDD-VD)/ID

.param WREF=0.390625u
.param LREF=1u
.param WAMP=0.390625u
.param LAMP=1u

VDD vdd 0 {VDD}

IREF vdd diode {ID}

MREF diode diode 0 0 CMOSN W={WREF} L={LREF}

RD vdd drain {RDVAL}
MAMP drain diode 0 0 CMOSN W={WAMP} L={LAMP}

VIN diode gate_ac AC 1
RAC gate_ac 0 1T

.control

op

print v(diode)
print v(drain)
print @mref[id]
print @mamp[id]
print @mamp[gm]
print @mamp[gds]
print @mamp[vgs]
print @mamp[vds]

let ro = 1/@mamp[gds]
let rd = 58982400
let rout = (ro*rd)/(ro+rd)
let gain = @mamp[gm]*rout

print ro
print rd
print rout
print gain

ac dec 100 1 1G
plot db(v(drain)/v(diode))

.endc

.end
```

Minimising power consumption in a current mirror based cs amplifier

1) Power consumption = 900 uWatts
2) w/l ratio=50
3) vd~vdd/2=0.9

With these assumptions:

The three dc bias points are
1) Drain current Id=500 uA
2) Vgs=(obtained from current mirror circuit)
3) Vds=0.9

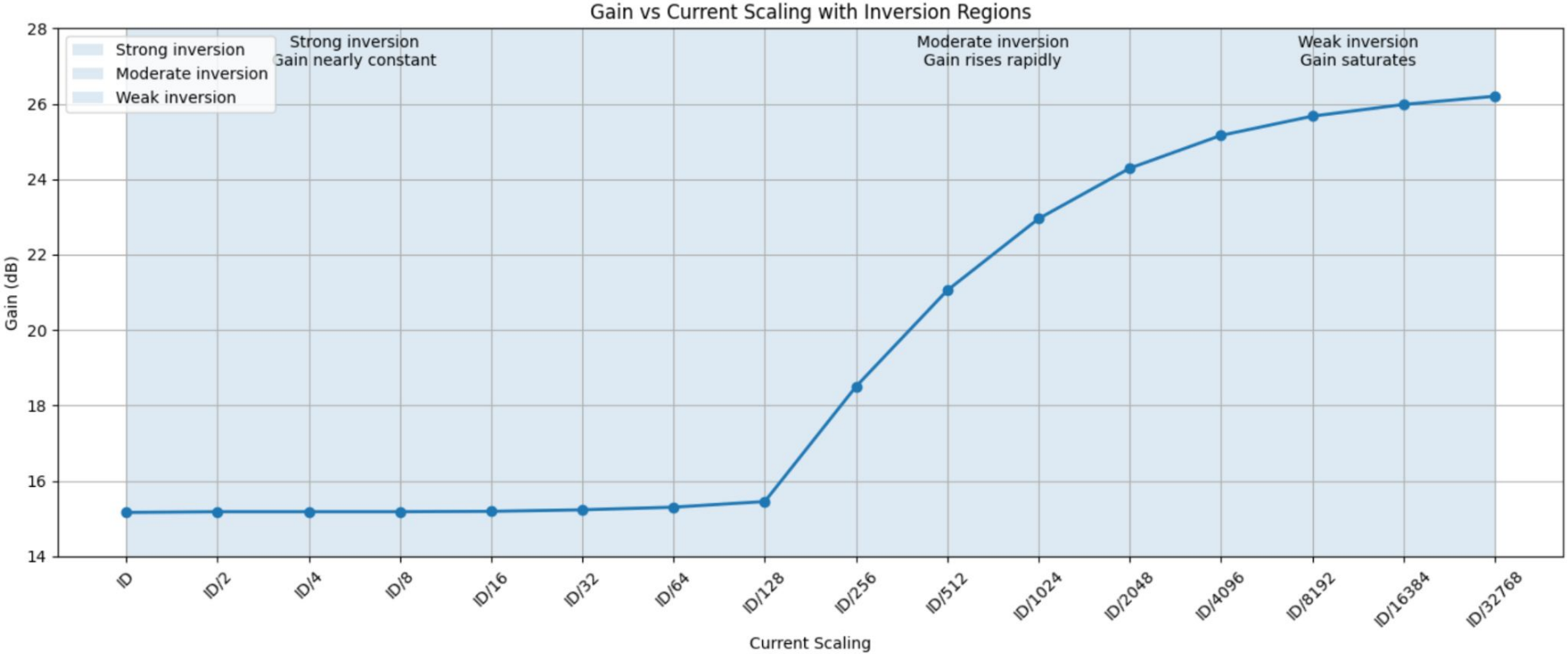
Results obtained from current, width and drain resistance scaling

Iteration	Width (um)	ID (uA)	VGS (V)	VDS (V)	gm (uS)	ro (MOhm)	RD (MOhn	gm/ID (V^-1)	Gain (V/V)	Gain (dB)
ID	50	504.74	0.702	0.892	3328.8	0.041	0.0018	6.6	5.73	15.16
ID/2	25	252.43	0.702	0.891	1664.5	0.082	0.0036	6.59	5.74	15.18
ID/4	12.5	126.23	0.701	0.895	832.4	0.163	0.0072	6.59	5.74	15.18
ID/8	6.25	63.12	0.701	0.89	416.4	0.326	0.0144	6.6	5.74	15.18
ID/16	3.125	31.56	0.7	0.891	208.6	0.648	0.0288	6.61	5.75	15.19
ID/32	1.5625	15.79	0.698	0.891	104.8	1.28	0.0576	6.64	5.77	15.23
ID/64	0.78125	7.9	0.695	0.89	52.87	2.5	0.1152	6.69	5.82	15.3
ID/128	0.390625	3.95	0.686	0.89	26.95	4.78	0.2304	6.82	5.92	15.45
ID/256	0.390625	1.99	0.603	0.881	19.43	7.26	0.4608	9.74	8.42	18.5
ID/512	0.390625	1.01	0.543	0.872	13.25	11.1	0.9216	13.12	11.28	21.05
ID/1024	0.390625	0.509	0.497	0.861	8.41	17.8	1.8432	16.52	14.05	22.95
ID/2048	0.390625	0.258	0.459	0.851	4.99	30	3.6864	19.35	16.39	24.29
ID/4096	0.390625	0.13	0.426	0.841	2.8	52.7	7.3728	21.53	18.11	25.16
ID/8192	0.390625	0.0656	0.395	0.832	1.5	96.3	14.7456	22.91	19.22	25.67
ID/16384	0.390625	0.0331	0.366	0.825	0.785	181	29.4912	23.72	19.92	25.98
ID/32768	0.390625	0.0167	0.337	0.818	0.405	346	58.9824	24.31	20.43	26.2

Observations and conclusions

1) When scaling is done till minimum width :
    a) gain values almost remains constant at 5.7 v/v
    b) all the dc bias points remain stable at calculated values
    c) the amplifier remains in strong inversion region
    d) the amplifier remains in saturation region

2) Below minimum width (id/256 and so on) only current is scaled :
    a) gain values start increasing rapidly
    b) all the dc bias points start drifting away from expected values
    c) Vgs reduces drastically and eventually becomes less than Vth
    d) the mosfet goes into weak inversion region at around id/4096
    e) the mosfet remains in saturation
    f) gm/id values increase rapidly with subsequent iterations


Gain vs Current Scaling with Inversion Regions
Strong inversion
Gain nearly constant
Moderate inversion
Gain rises rapidly
Weak inversion
Gain saturates
Strong inversion
Moderate inversion
Weak inversion
Gain (dB)
28
26
24
22
20
18
16
14
ID
ID/2
ID/4
ID/8
ID/16
ID/32
ID/64
ID/128
ID/256
ID/512
ID/1024
ID/2048
ID/4096
ID/8192
ID/16384
ID/32768
Current Scaling

In Weak inversion region (Vgs<Vth):

1)gm/id saturates to a constant value of i/(n*Vt)

2)halving current results in a constant delta in Vgs because of exponential relation

3)simulation values support these facts

CS amplifier power consumption optimisation, automated using python scripts

1.Using python scripts , we automate the process of scaling Rd, Id and W/l ratio.

2. The scaling is done until either W < Wmin or gain exceeds a certain error threshold

3. This ensures that we find the minimum power required to produce the required gain. Hence optimising the cs amplifier for power consumption for a given model of mosfet.

Results

```
Current Mirror Biased CS Amplifier Automation
Press Enter to use default values.

Enter VDD in volts [default 1.8]:
Enter initial power in watts [default 900e-6]:
Enter channel length L in meters [default 1e-6]:
Enter minimum width Wmin in meters [default 0.390625e-6]:
Enter initial W/L ratio [default 50]:
Enter target drain voltage [default VDD/2]:
Enter gain tolerance percentage [default 5]:
Enter max scale factor [default 32768]:

Initial calculated values:
ID_initial = 5.000000e-04 A
RD_initial = 1800.00 ohm
W_initial = 5.000000e-05 m
L = 1.000000e-06 m

      ID | W mode:       scaled | W= 50.0000 um | ID= 504.7429 uA | Gain=   5.738 V/V | Error=  0.00%
    ID/2 | W mode:       scaled | W= 25.0000 um | ID= 252.4272 uA | Gain=   5.739 V/V | Error=  0.01%
    ID/4 | W mode:       scaled | W= 12.5000 um | ID= 126.2293 uA | Gain=   5.740 V/V | Error=  0.02%
    ID/8 | W mode:       scaled | W=  6.2500 um | ID=  63.1202 uA | Gain=   5.742 V/V | Error=  0.08%
   ID/16 | W mode:       scaled | W=  3.1250 um | ID=  31.5638 uA | Gain=   5.752 V/V | Error=  0.23%
   ID/32 | W mode:       scaled | W=  1.5625 um | ID=  15.7852 uA | Gain=   5.774 V/V | Error=  0.63%
   ID/64 | W mode:       scaled | W=  0.7812 um | ID=   7.8960 uA | Gain=   5.823 V/V | Error=  1.48%
  ID/128 | W mode:fixed_at_Wmin | W=  0.3906 um | ID=   3.9515 uA | Gain=   5.923 V/V | Error=  3.21%
  ID/256 | W mode:fixed_at_Wmin | W=  0.3906 um | ID=   1.9937 uA | Gain=   8.420 V/V | Error= 46.73%

Stopping: gain error exceeded 5% at ID/256

Done.
CSV saved to: C:\Users\kisha\Downloads\ngspice-46_64\Spice64\bin\automation_results.csv
Plots saved to: C:\Users\kisha\Downloads\ngspice-46_64\Spice64\bin
```

Designing a amplifier with a target gain , while optimizing power consumption

1) Target gain = av= gm*rd
2) gm = sqrt(2*un*cox*(w/l)*Id)
3) An initial W/l ratio is found out for the required gain instead of assuming it to be an arbitrary value like 50
4) Now using a python script, we perform binary search on the optimum w/l ratio to obtain the required gain.Because, BSIM models do not follow square laws , hence finding accurate w/l ratio via hand calculations seems to not be feasible.
5) Now we arrive at the correct w/l ratio
6) We can start scaling the current down to optimise for power.

Python script

```
Enter VDD in volts [default 1.8]:
Enter initial power in watts [default 900e-6]:
Enter target gain in V/V [default 10]:
Enter channel length L in meters [default 1e-6]:
Enter minimum width Wmin in meters [default 0.390625e-6]: 0.27e-6
Enter μnCox in uA/V^2 [default 230]:
Enter target drain voltage [default VDD/2]:
Enter gain tolerance percentage [default 5]:
Enter initial W/L refinement tolerance percentage [default 1]:
Enter max scale factor [default 32768]:

Initial hand-calculated values:
ID_initial = 5.000000e-04 A
RD_initial = 1800.00 ohm
Target gain = 10.000 V/V
Required gm ≈ 5.555556e-03 S
Estimated W/L = 134.192
Estimated W = 1.341922e-04 m
L = 1.000000e-06 m
Wmin = 2.700000e-07 m

Refining initial W/L for target gain...

Refine 01 | W= 134.1922 um | W/L=  134.192 | Gain=   9.400 V/V | Error=  6.00%
Refine 02 | W= 151.8630 um | W/L=  151.863 | Gain=   9.915 V/V | Error=  0.85%

Final initial design:
W_initial = 1.518630e-04 m
W/L_initial = 151.863
Initial simulated gain = 9.915 V/V
Initial gain error vs target = 0.85%
```

Final results obtained after optimisation ,achieving a target gain while reducing power consumption

```
      ID | W mode:       scaled | W= 151.8630 um | W/L=   151.86 | ID=  511.3193 uA | Gain=   9.915 V/V | Error=  0.85%
    ID/2 | W mode:       scaled | W=  75.9315 um | W/L=    75.93 | ID=  255.8254 uA | Gain=   9.918 V/V | Error=  0.82%
    ID/4 | W mode:       scaled | W=  37.9658 um | W/L=    37.97 | ID=  127.9565 uA | Gain=   9.919 V/V | Error=  0.81%
    ID/8 | W mode:       scaled | W=  18.9829 um | W/L=    18.98 | ID=   63.9904 uA | Gain=   9.919 V/V | Error=  0.81%
   ID/16 | W mode:       scaled | W=   9.4914 um | W/L=     9.49 | ID=   31.9995 uA | Gain=   9.920 V/V | Error=  0.80%
   ID/32 | W mode:       scaled | W=   4.7457 um | W/L=     4.75 | ID=   16.0021 uA | Gain=   9.925 V/V | Error=  0.75%
   ID/64 | W mode:       scaled | W=   2.3729 um | W/L=     2.37 | ID=    8.0031 uA | Gain=   9.940 V/V | Error=  0.60%
  ID/128 | W mode:       scaled | W=   1.1864 um | W/L=     1.19 | ID=    4.0036 uA | Gain=   9.975 V/V | Error=  0.25%
  ID/256 | W mode:       scaled | W=   0.5932 um | W/L=     0.59 | ID=    2.0039 uA | Gain=  10.049 V/V | Error=  0.49%
  ID/512 | W mode:       scaled | W=   0.2966 um | W/L=     0.30 | ID=    1.0041 uA | Gain=  10.195 V/V | Error=  1.95%
 ID/1024 | W mode:fixed_at_Wmin | W=   0.2700 um | W/L=     0.27 | ID=    0.5072 uA | Gain=  12.692 V/V | Error= 26.92%

Stopping: gain error exceeded 5% at ID/1024
```

1) Power reduced by 512 times
2) Target gain of 10 achieved within 2% error
3) All physical limitations such as Wmin, Lmin are taken into account.

Scaling using .ac output as reference instead of gain calculated using DC bias pts

```
ID_initial = 5.000000e-04 A
RD_initial = 1800.00 ohm
Target AC gain = 10.000 V/V
Required gm ≈ 5.555556e-03 S
Estimated W/L = 134.192
Estimated W = 1.341922e-04 m
L = 1.000000e-06 m
Wmin = 3.906250e-07 m

Refining initial W/L for target AC gain...

Refine 01 | W= 134.1922 um | W/L=  134.192 | AC Gain=   8.993 V/V | Pred Gain=   9.400 V/V | Error= 10.07%
Refine 02 | W= 165.9156 um | W/L=  165.916 | AC Gain=   9.801 V/V | Pred Gain=  10.286 V/V | Error=  1.99%
Refine 03 | W= 172.7283 um | W/L=  172.728 | AC Gain=   9.954 V/V | Pred Gain=  10.456 V/V | Error=  0.46%

Final initial design:
W_initial = 1.727283e-04 m
W/L_initial = 172.728
Initial AC gain = 9.954 V/V
Initial predicted gain = 10.456 V/V
Initial gain error vs target = 0.46%

      ID | W mode:       scaled | W= 172.7283 um | ID=  512.1731 uA | Pred=  10.456 | AC=   9.954 | AC-target err=  0.46% | Pred-AC err=  4.79%
    ID/2 | W mode:       scaled | W=  86.3641 um | ID=  256.2714 uA | Pred=  10.459 | AC=  10.202 | AC-target err=  2.02% | Pred-AC err=  2.46%
    ID/4 | W mode:       scaled | W=  43.1821 um | ID=  128.1845 uA | Pred=  10.460 | AC=  10.330 | AC-target err=  3.30% | Pred-AC err=  1.24%
    ID/8 | W mode:       scaled | W=  21.5910 um | ID=   64.1056 uA | Pred=  10.461 | AC=  10.395 | AC-target err=  3.95% | Pred-AC err=  0.63%
   ID/16 | W mode:       scaled | W=  10.7955 um | ID=   32.0573 uA | Pred=  10.461 | AC=  10.429 | AC-target err=  4.29% | Pred-AC err=  0.31%
   ID/32 | W mode:       scaled | W=   5.3978 um | ID=   16.0310 uA | Pred=  10.465 | AC=  10.448 | AC-target err=  4.48% | Pred-AC err=  0.16%
   ID/64 | W mode:       scaled | W=   2.6989 um | ID=    8.0174 uA | Pred=  10.477 | AC=  10.469 | AC-target err=  4.69% | Pred-AC err=  0.08%
  ID/128 | W mode:       scaled | W=   1.3494 um | ID=    4.0106 uA | Pred=  10.507 | AC=  10.502 | AC-target err=  5.02% | Pred-AC err=  0.04%

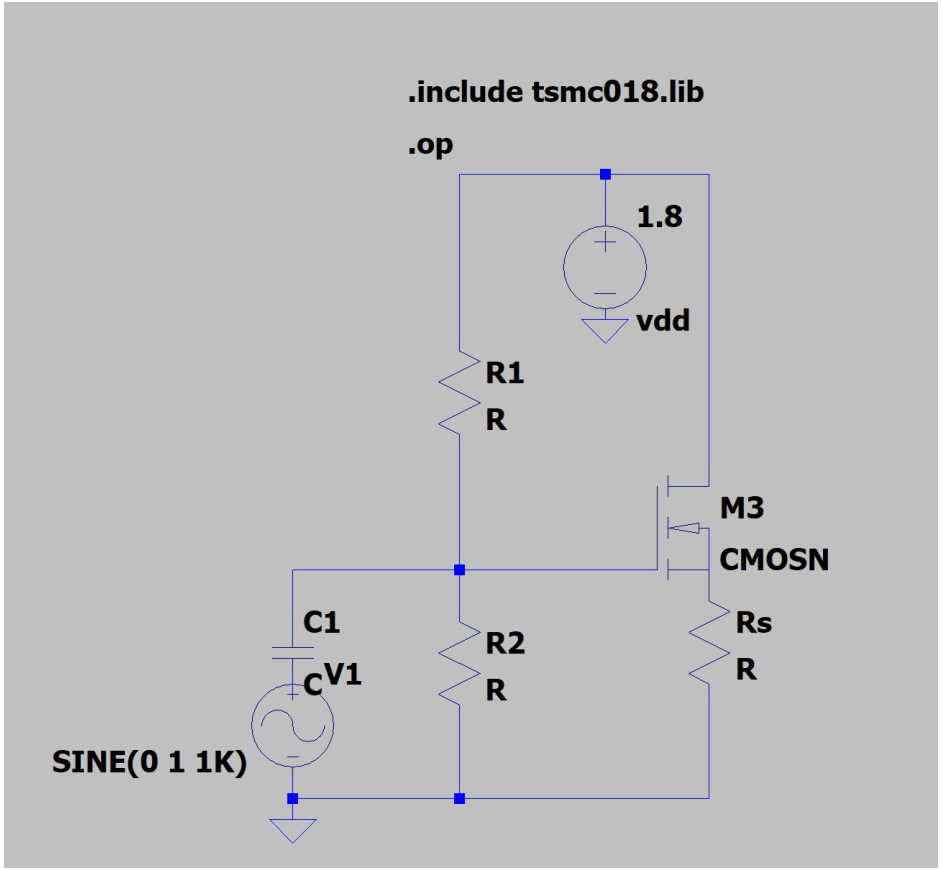
Stopping: AC gain error exceeded 5% at ID/128
```

Observations and conclusions

1) when using .ac gain , the scaling stops at id/64 ,a 4x reduction compared to using gain calculated using dc bias pts obtained from . op analysis

2)the error between AC gain and predicted gain reduces drastically as we scale down , because of reduction in parasitic capacitances and other irregularities , resembling a ideal model at lower currents

Common drain amplifier automated using python scripts



Calculations

- A common-drain amplifier was designed with VDD = 1.8 V, source voltage VS = 0.9 V and power = 900 μW. This gave ID = 500 μA and RS = VS/ID = 1.8 kΩ.
- Using Av = gmRS/(1 + gmRS) with a target gain of 0.95, gm was calculated as 10.56 mS. The squarelaw equation gave W/L = 484.4, so W = 87.2 μm for L = 0.18 μm.
- The hand-calculated gate voltage did not produce VS = 0.9 V because the assumed threshold voltage did not include body effect. Therefore, VG was swept in SPICE until VS reached 0.9 V.
- The required gate voltage was 1.6473 V. A divider of RTOP = 92.67 kΩ and RBOTTOM = 1 MΩ was used to generate it.
- Current was repeatedly halved. At each step, RS was doubled and W was halved, while VG was recalculated. This kept the gain nearly constant at about 0.78.

- Output impedance approximately doubled at every step. It was used as the stopping condition because a source follower must drive the next stage without excessive voltage loss.
- Assuming the next-stage input resistance was 100 kΩ and requiring at least 95% voltage transfer, the maximum acceptable Rout was 5.263 kΩ.
- ID/64 was the final valid point, with ID = 7.8125 μA, power = 14.06 μW, gain = 0.777 and Rout = 5.223 kΩ. ID/128 was rejected because Rout increased to 10.15 kΩ.

Python automation results

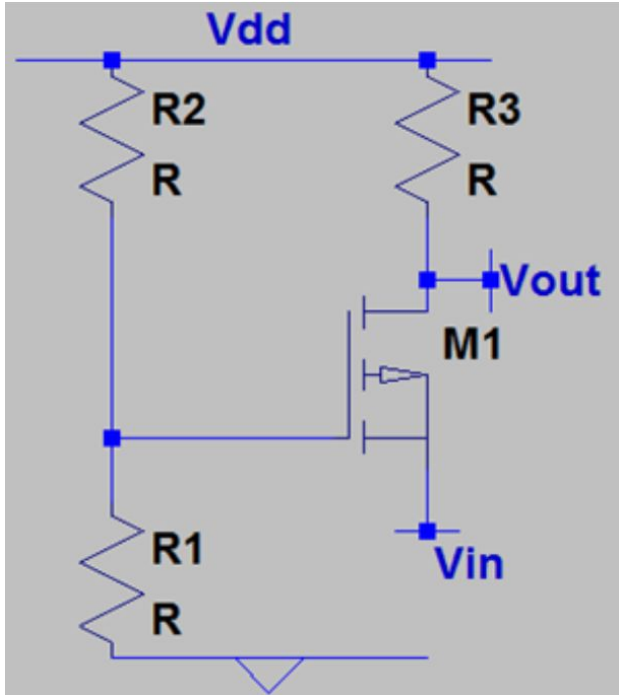
```
Model: C:\Users\kisha\Downloads\ngspice-46_64\Spice64\bin\tsmc018.lib
Output folder: C:\Users\kisha\Downloads\ngspice-46_64\Spice64\bin\cd_scaling_results
Rout limit: 5263.158 ohm
Equivalent transfer rule: at least 95.0% into 100000.000 ohm

[    ID] ID=  500.0000 uA, RS=    1.8000 kohm, W=   87.2000 um
        VG=1.647338 V, gain=0.78098, Rout=90.245 ohm, transfer=99.910%, sat.margin=0.824091 V
[  ID/2] ID=  250.0000 uA, RS=    3.6000 kohm, W=   43.6000 um
        VG=1.645066 V, gain=0.78090, Rout=174.220 ohm, transfer=99.826%, sat.margin=0.827505 V
[  ID/4] ID=  125.0000 uA, RS=    7.2000 kohm, W=   21.8000 um
        VG=1.643648 V, gain=0.78077, Rout=341.910 ohm, transfer=99.659%, sat.margin=0.829263 V
[  ID/8] ID=   62.5000 uA, RS=   14.4000 kohm, W=   10.9000 um
        VG=1.642369 V, gain=0.78053, Rout=676.291 ohm, transfer=99.328%, sat.margin=0.830232 V
[ ID/16] ID=   31.2500 uA, RS=   28.8000 kohm, W=    5.4500 um
        VG=1.640575 V, gain=0.78006, Rout=1341.056 ohm, transfer=98.677%, sat.margin=0.830873 V
[ ID/32] ID=   15.6250 uA, RS=   57.6000 kohm, W=    2.7250 um
        VG=1.637373 V, gain=0.77911, Rout=2654.860 ohm, transfer=97.414%, sat.margin=0.831453 V
[ ID/64] ID=    7.8125 uA, RS=  115.2000 kohm, W=    1.3625 um
        VG=1.631277 V, gain=0.77716, Rout=5223.353 ohm, transfer=95.036%, sat.margin=0.832159 V
[ID/128] ID=    3.9062 uA, RS=  230.4000 kohm, W=    0.6813 um
        VG=1.619740 V, gain=0.77308, Rout=10152.967 ohm, transfer=90.783%, sat.margin=0.833082 V
  STOP: Rout 10152.967 ohm exceeded limit 5263.158 ohm

Completed.
Stop reason: Rout 10152.967 ohm exceeded limit 5263.158 ohm
CSV: C:\Users\kisha\Downloads\ngspice-46_64\Spice64\bin\cd_scaling_results\cd_scaling_results.csv
Summary: C:\Users\kisha\Downloads\ngspice-46_64\Spice64\bin\cd_scaling_results\summary.txt
Each step folder also contains its generated netlists and logs.

C:\Users\kisha>
```

Common Gain amplifier circuit



Calculations

- A common-gate amplifier was designed with VDD = 1.8 V, drain voltage VD = 0.9 V and power = 900 μW. This gave ID = 500 μA and RD = (VDD − VD)/ID = 1.8 kΩ.
- Using Av ≈ gmRD with a target gain of 10, gm was calculated as 5.56 mS. The square-law equation gave W/L = 134.2, so W = 24.15 μm for L = 0.18 μm.
- The hand-calculated gate voltage of 0.546 V did not produce ID = 500 μA because the squarelaw approximation did not accurately represent the BSIM model. Therefore, VG was swept in SPICE until VD reached 0.9 V.
- The required gate voltage was 0.6355 V. A divider of RTOP = 1.832 MΩ and RBOTTOM = 1 MΩ was used to generate it, with a capacitor connected from the gate to ground to make it an AC ground.
- The baseline simulated gain was 10.15, with Rin = 177.3 Ω and Rout = 1.419 kΩ.
- Current was repeatedly halved. At each step, RD was doubled, W was halved and VG was recalculated to maintain VD = 0.9 V.
- The gain gradually increased because gm, gmb and gds did not scale exactly according to the ideal squarelaw model. Rin and Rout approximately doubled at every step.
- With a target gain tolerance of ±5%, ID/4 was the first failing case with a gain of 10.51. Therefore, ID/2 was the final accepted case, with ID = 250 μA, gain = 10.37 and power = 450 μW.

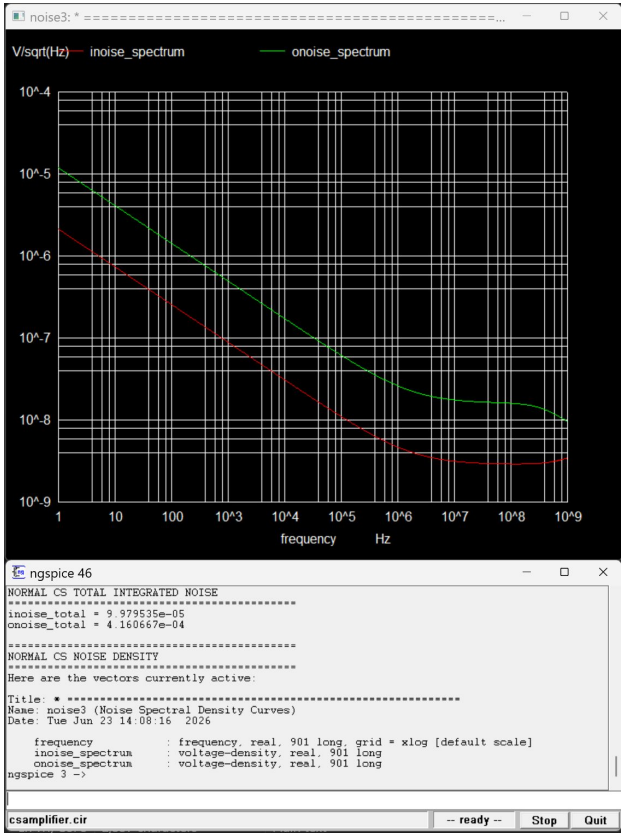
Python Automation Output (scaling stops at Id/2 as gain error exceeds target)

```
Output folder: C:\Users\kisha\Downloads\ngspice-46_64\Spice64\bin\cg_scaling_results_v4

[    ID] ID=  500.0000 uA, RD=    1.8000 kohm, W=   24.1500 um
        VG=0.635496 V, VD=0.900000 V, ID=500.0000 uA
        gain=10.152070, Rin=177.304 ohm, Rout=1419.101 ohm, sat.margin=0.787330 V
[  ID/2] ID=  250.0000 uA, RD=    3.6000 kohm, W=   12.0750 um
        VG=0.632182 V, VD=0.900000 V, ID=250.0000 uA
        gain=10.371540, Rin=347.104 ohm, Rout=2821.056 ohm, sat.margin=0.791066 V
[  ID/4] ID=  125.0000 uA, RD=    7.2000 kohm, W=    6.0375 um
        VG=0.628783 V, VD=0.900000 V, ID=125.0000 uA
        gain=10.510130, Rin=685.053 ohm, Rout=5615.530 ohm, sat.margin=0.793408 V
  STOP: Gain error from target 5.101% exceeded 5.000%

Completed.
Stop reason: Gain error from target 5.101% exceeded 5.000%
```

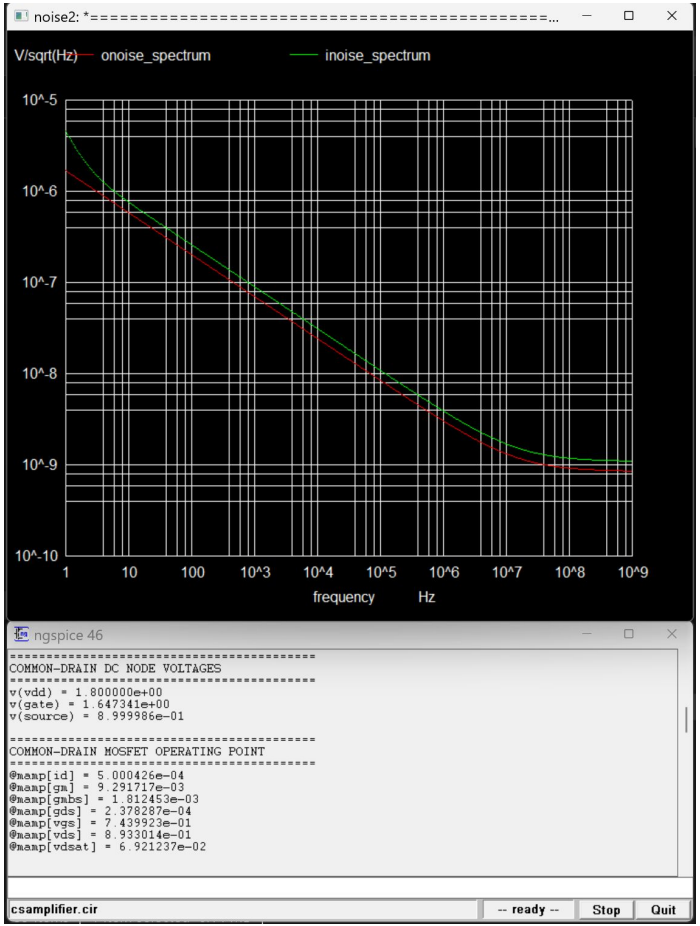
Noise analysis : (plotting input referred noise for cs amplifier) (1Hz to 1Ghz)



Observed results

1) Innoise refers to the output noise referred to the input , and equals 9.97e-5 v
2) Onnoise refers to the total output noise at the drain , and equals 4.16e-4 v

Noise analysis : (plotting input referred noise for cd amplifier) (1Hz to 1Ghz)




```
==========
COMMON-DRAIN AC RESULTS AT 1 kHz
==========
mag(av_ac) = 7.781565e-01
db(av_ac) = -2.17866e+00
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.

No. of Data Rows : 901
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.

No. of Data Rows : 901

No. of Data Rows : 1

==========
COMMON-DRAIN TOTAL INTEGRATED NOISE
==========
inoise_total = 3.804836e-05
onoise_total = 2.952083e-05

==========
COMMON-DRAIN NOISE DENSITY
==========
Here are the vectors currently active:

Title: *=============================================================
Name: noise2 (Noise Spectral Density Curves)
Date: Tue Jun 23 14:36:01  2026

    frequency           : frequency, real, 901 long, grid = xlog [default scale]
    inoise_spectrum     : voltage-density, real, 901 long
    onoise_spectrum     : voltage-density, real, 901 long
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.

Initial Transient Solution
--------------------------

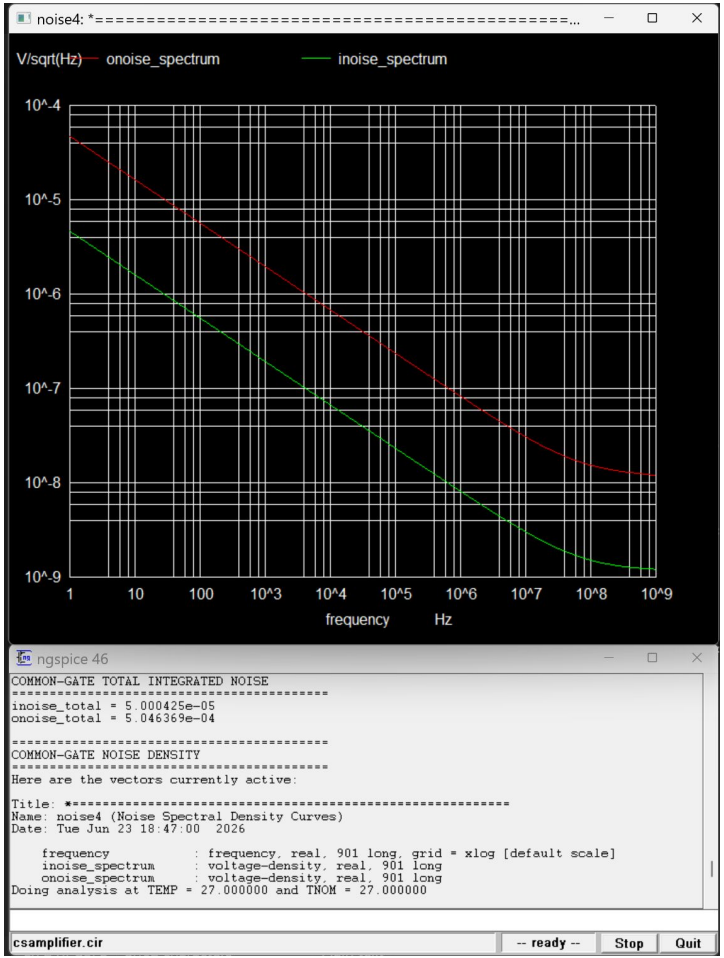
Node                                   Voltage
----                                   -------
vdd                                        1.8
gate                                   1.64734
vin                                          0
source                                0.899999
vin#branch                                   0
vdd#branch                        -0.000501647


No. of Data Rows : 5008
ngspice 2 ->
```

Common drain results

- inoise refers to the total output noise mathematically referred back to the input and equals 3.8×10^{-5} V RMS.
- Onoise refers to the total output noise measured at the source node and equals 2.9×10^{-5} V RMS.

Noise analysis : (plotting input referred noise for cg amplifier) (1Hz to 1Ghz)



Inoise total refers to the total drain-output noise mathematically referred back to the source input and equals 5.000425×10^{-5} V RMS.

Onoise total refers to the total noise actually present at the drain output and equals 5.046369×10^{-4} V RMS.

3db bandwidth calculated for cs amplifier for every iteration

```
      ID | W mode:       scaled | W= 172.7505 um | ID=  512.1740 uA | Pred=  10.456 | AC=   9.954 | fL=   0.1591 Hz | fH=     389 MHz | BW=     389 MHz
r=  0.46% | Pred-AC err=  4.80%
    ID/2 | W mode:       scaled | W=  86.3752 um | ID=  256.2718 uA | Pred=  10.459 | AC=  10.202 | fL=   0.1591 Hz | fH=   455.1 MHz | BW=   455.1 MHz
r=  2.02% | Pred-AC err=  2.46%
    ID/4 | W mode:       scaled | W=  43.1876 um | ID=  128.1847 uA | Pred=  10.461 | AC=  10.330 | fL=    0.159 Hz | fH=   495.8 MHz | BW=   495.8 MHz
r=  3.30% | Pred-AC err=  1.25%
    ID/8 | W mode:       scaled | W=  21.5938 um | ID=   64.1057 uA | Pred=  10.461 | AC=  10.395 | fL=   0.1589 Hz | fH=   518.1 MHz | BW=   518.1 MHz
r=  3.95% | Pred-AC err=  0.63%
   ID/16 | W mode:       scaled | W=  10.7969 um | ID=   32.0573 uA | Pred=  10.462 | AC=  10.429 | fL=   0.1587 Hz | fH=   529.8 MHz | BW=   529.8 MHz
r=  4.29% | Pred-AC err=  0.32%
   ID/32 | W mode:       scaled | W=   5.3985 um | ID=   16.0310 uA | Pred=  10.465 | AC=  10.448 | fL=   0.1583 Hz | fH=   535.8 MHz | BW=   535.8 MHz
r=  4.48% | Pred-AC err=  0.16%
   ID/64 | W mode:       scaled | W=   2.6992 um | ID=    8.0174 uA | Pred=  10.477 | AC=  10.469 | fL=   0.1575 Hz | fH=   538.9 MHz | BW=   538.9 MHz
r=  4.69% | Pred-AC err=  0.08%
  ID/128 | W mode:       scaled | W=   1.3496 um | ID=    4.0106 uA | Pred=  10.507 | AC=  10.502 | fL=   0.1559 Hz | fH=   540.7 MHz | BW=   540.7 MHz
r=  5.02% | Pred-AC err=  0.04%
```

Observations

1. The upper cutoff frequency increases as current and transistor width are reduced because the smaller MOSFET has lower parasitic capacitances.
2. The lower cutoff frequency remains almost constant near 0.159 Hz because it is mainly determined by the fixed 1 MΩ bias resistance and 1 μF input capacitor.
3. Since the lower cutoff frequency is extremely small compared with the upper cutoff frequency, the 3 dB bandwidth is practically equal to the upper cutoff frequency.
4. The gain remains close to the target value of 10 because the drain resistance is increased as the current and transistor width are reduced.
5. The ID/128 case has about 5.02% gain error and is the first case outside the 5% tolerance, making ID/64 the last acceptable case.

3db bandwidth calculated for cd amplifier for multiple iterations

```
Enter VDD in volts [default 1.8]:
Enter target AC gain in V/V [default 0.95]:
Enter channel length L in meters [default 0.18e-6]:
Enter starting width W in meters [default 87.2e-6]:
Enter minimum width Wmin in meters [default 0.390625e-6]:
Enter starting RS value in ohms [default 1800]:
Enter starting RTOP value in ohms [default 92.67k]:
Enter starting RBOTTOM value in ohms [default 1Meg]:
Enter gain tolerance percentage [default 5]:
Enter initial width refinement tolerance % [default 1]: 3
Enter midband measurement frequency in Hz [default 1000]:
Enter AC sweep start frequency in Hz [default 1]:
Enter AC sweep stop frequency in Hz [default 1e9]:
Enter AC sweep points per decade [default 100]:
Enter max scale factor [default 16]:

Refining initial W/L for target CD AC gain...

Refine 01 | W=  87.2000 um | W/L=  484.444 | AC Gain=  0.9218 V/V | Pred Gain=  0.9260 V/V | Error=  2.96%

Final initial design:
W_initial = 8.720000e-05 m
W/L_initial = 484.444
Initial AC gain = 0.9218 V/V
Initial predicted gain = 0.9260 V/V
Initial gain error vs target = 2.96%

      ID | W mode:       scaled | W=  87.2000 um | ID=  600.7979 uA | Pred=  0.9260 | AC=  0.9218 | fL=    1.877 Hz | fH=   not found | BW=   not found
r=  2.96%
    ID/2 | W mode:       scaled | W=  43.6000 um | ID=  301.1408 uA | Pred=  0.9261 | AC=  0.9240 | fL=   not found | fH=   not found | BW=   not found
r=  2.74%
    ID/4 | W mode:       scaled | W=  21.8000 um | ID=  150.8089 uA | Pred=  0.9261 | AC=  0.9251 | fL=   not found | fH=   not found | BW=   not found
r=  2.62%
    ID/8 | W mode:       scaled | W=  10.9000 um | ID=   75.5174 uA | Pred=  0.9261 | AC=  0.9256 | fL=   not found | fH=   not found | BW=   not found
r=  2.57%
   ID/16 | W mode:       scaled | W=   5.4500 um | ID=   37.8410 uA | Pred=  0.9262 | AC=  0.9259 | fL=   not found | fH=   not found | BW=   not found
r=  2.54%
```

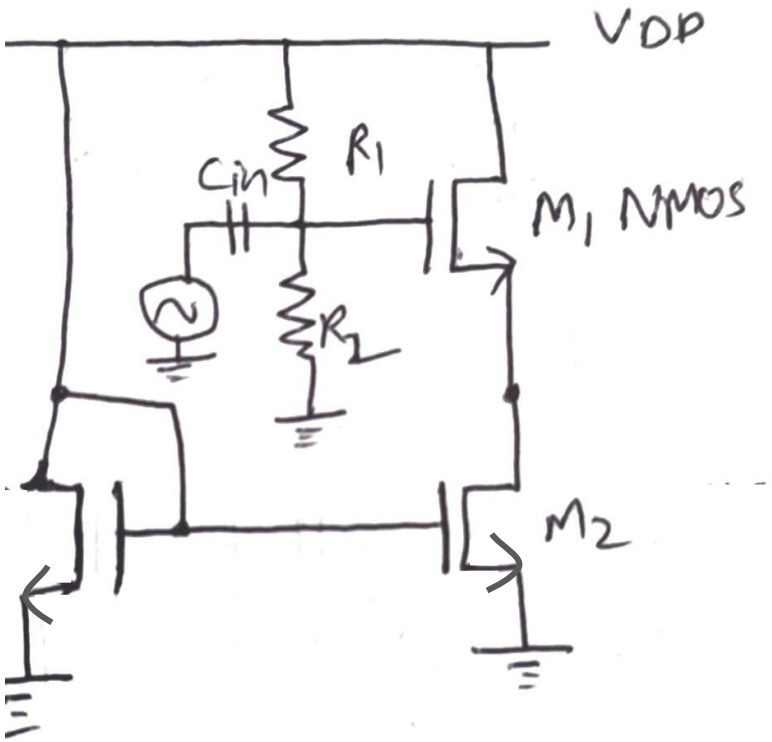
Observations

1. The lower cutoff and upper cutoff frequencies go out of bounds for the given bandwidth.

Closed form expression relating noise of amplifiers with device parameters

$$In^2/\triangle f = (4KT/(Rd*gm^2)) + (8KT/(3*gm)) + k / (W*L*Cox*f)$$

Common Drain Amplifier with a NMOS Active Load (current mirror biasing)



$$A_v = \frac{g_m[R_p \| R_n]}{g_m[R_p \| R_n]+1}$$

$$A_V = \frac{g_m R_n}{1+g_m R_n}$$

$$A_V = \frac{g_m}{\left(1+g_m\frac{1}{g_{dsn}}\right)g_{dsn}}$$

$$A_V = \left[\frac{g_m}{g_{dsn}+g_m}\right]$$

$$\left[\frac{g_m}{g_{dsn}} = \frac{A_V}{1-A_V}\right]$$

- V_{DD}, P, $A_V = 0.97$

$\frac{g_m}{g_{ds}} = 32.33$, from V_{gs} sweep from (0 to 1.8)V

we get

for $\frac{g_m}{g_{ds}} = 32.5$, $V_{GS} = 0.7$V

$I_D/W = 2$ mA/μm

W = obtained

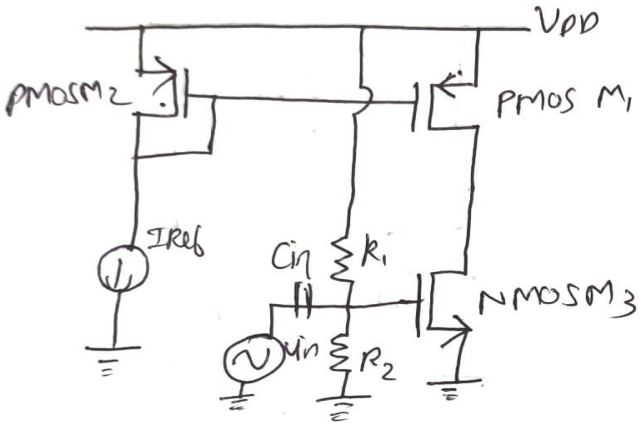
Choosing $V_S = 0.9$, $V_G = V_S + V_{GS}$ = 10 μm

Lookup table for gm/gds reference

VGS	ID	GM	GDS	RO	VTH	VOV	GM_ID	ID_W	GM_GDS
0	5.81E-11	1.57E-09	3.73E-11	2.68E+10	0.495193	-0.49519	27.00939	5.81E-06	42.09993
0.001	5.97E-11	1.61E-09	3.83E-11	2.61E+10	0.495193	-0.49419	27.02044	5.97E-06	42.09961
0.002	6.13E-11	1.66E-09	3.94E-11	2.54E+10	0.495193	-0.49319	27.03119	6.13E-06	42.09929
0.003	6.30E-11	1.70E-09	4.05E-11	2.47E+10	0.495193	-0.49219	27.04164	6.30E-06	42.09896
0.004	6.47E-11	1.75E-09	4.16E-11	2.41E+10	0.495193	-0.49119	27.05181	6.47E-06	42.09863
0.005	6.65E-11	1.80E-09	4.27E-11	2.34E+10	0.495193	-0.49019	27.0617	6.65E-06	42.09829
0.006	6.83E-11	1.85E-09	4.39E-11	2.28E+10	0.495193	-0.48919	27.07131	6.83E-06	42.09795
0.007	7.02E-11	1.90E-09	4.51E-11	2.22E+10	0.495193	-0.48819	27.08066	7.02E-06	42.0976
0.008	7.21E-11	1.95E-09	4.64E-11	2.16E+10	0.495193	-0.48719	27.08975	7.21E-06	42.09725
0.009	7.41E-11	2.01E-09	4.77E-11	2.1E+10	0.495193	-0.48619	27.09859	7.41E-06	42.0969
0.01	7.61E-11	2.06E-09	4.90E-11	2.04E+10	0.495193	-0.48519	27.10718	7.61E-06	42.09654
0.011	7.82E-11	2.12E-09	5.04E-11	1.99E+10	0.495193	-0.48419	27.11553	7.82E-06	42.09617
0.012	8.04E-11	2.18E-09	5.18E-11	1.93E+10	0.495193	-0.48319	27.12365	8.04E-06	42.0958
0.013	8.26E-11	2.24E-09	5.32E-11	1.88E+10	0.495193	-0.48219	27.13154	8.26E-06	42.09542
0.014	8.48E-11	2.30E-09	5.47E-11	1.83E+10	0.495193	-0.48119	27.1392	8.48E-06	42.09504
0.015	8.72E-11	2.37E-09	5.62E-11	1.78E+10	0.495193	-0.48019	27.14665	8.72E-06	42.09466
0.016	8.96E-11	2.43E-09	5.78E-11	1.73E+10	0.495193	-0.47919	27.15389	8.96E-06	42.09427
0.017	9.20E-11	2.50E-09	5.94E-11	1.68E+10	0.495193	-0.47819	27.16092	9.20E-06	42.09387
0.018	9.46E-11	2.57E-09	6.10E-11	1.64E+10	0.495193	-0.47719	27.16775	9.46E-06	42.09347
0.019	9.72E-11	2.64E-09	6.27E-11	1.59E+10	0.495193	-0.47619	27.17438	9.72E-06	42.09306
0.02	9.99E-11	2.71E-09	6.45E-11	1.55E+10	0.495193	-0.47519	27.18083	9.99E-06	42.09265
0.021	1.03E-10	2.79E-09	6.63E-11	1.51E+10	0.495193	-0.47419	27.18708	1.03E-05	42.09223
0.022	1.05E-10	2.87E-09	6.81E-11	1.47E+10	0.495193	-0.47319	27.19316	1.05E-05	42.0918
0.023	1.08E-10	2.95E-09	7.00E-11	1.43E+10	0.495193	-0.47219	27.19906	1.08E-05	42.09137
0.024	1.11E-10	3.03E-09	7.20E-11	1.39E+10	0.495193	-0.47119	27.20478	1.11E-05	42.09094

Common source Amplifier with active PMOS load (current mirror biasing)

(CS)



$A_V = g_m(R_{on} \| R_{op})$

$A_V = \frac{g_m(R_{on})}{2R_{on}}$

$A_V = \frac{g_m R_{on}}{2}$

$A_V = \frac{g_m}{g_{ds}(2)}$

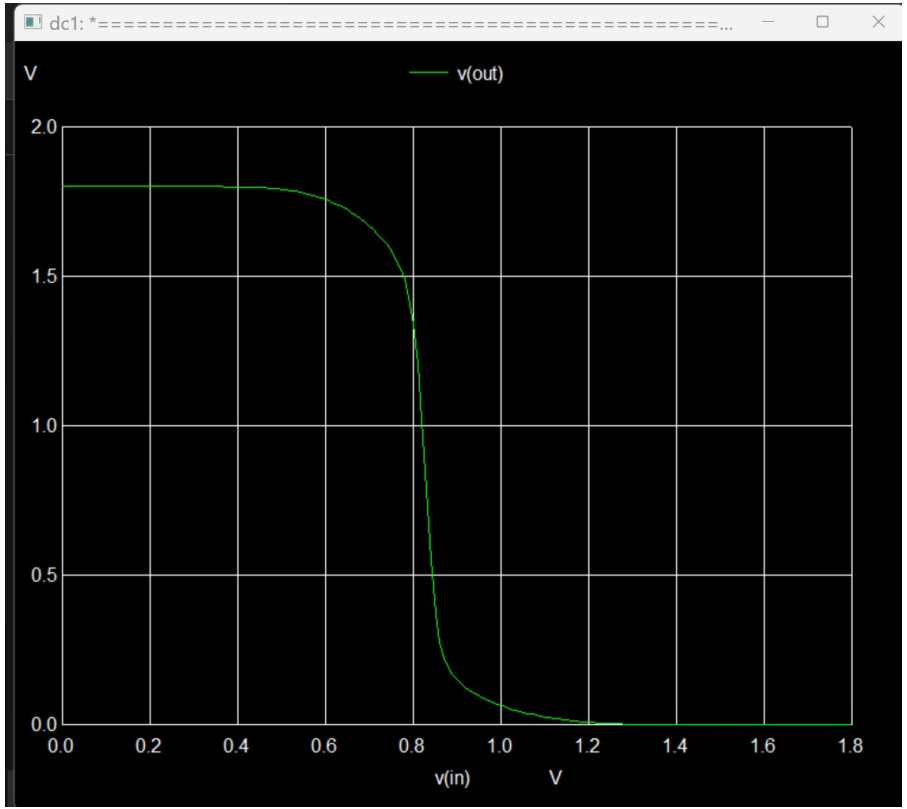
$\left[\frac{g_m}{g_{ds}} = 2A_V\right]$

∘) let Target gain be,

$\frac{g_m}{g_{ds}} = 2(10)$

$\left[\frac{g_m}{g_{ds}} = 20\right]$ from lookup table for that (Nmos)

Small signal analysis of CMOS Inverter



1) Gain = -(gm1+gm2)/(gds1+gds2)

2) vout=vin at 0.82 volts

3) Hence dc bias can be set at 0.82 volts to obtain amplifier behaviour

CD Amplifier with active load scaled with additional constraints(noise,bandwidth)

```
--- 1. Technology parameters ---
Enter Supply Voltage VDD (V) [default 1.8]:
Enter Approx Threshold Voltage VTH (V) [default 0.45]: 0.3666
Enter process un*cox (uA/V^2) [default 230]:
Enter Input Transistor Length L2 (um) [default 0.18]:
Enter Load Transistor Length L1 (um) [default 0.54]: 0.18
Enter Minimum Allowed Width Wmin (um) [default 0.27]:

--- 2. Circuit Specifications & Constraints ---
Enter Target Voltage Gain (V/V) [default 0.90]:
Enter Minimum Bandwidth Spec (MHz) [default 100]:
Enter Maximum Integrated Input Noise (uV) [default 100]:
Enter Maximum Current Budget (mA) [default 1.0]:
Enter Output Load Resistance RL (Ohm) [default 1000]:
Enter Output Load Capacitance CL (pF) [default 2.0]:
Enter Noise Suppression Ratio alpha (gm1/gm2) [default 0.5]:

--- 3. Procedural Baseline Calculations ---
 > gm bound from Gain Spec:      9.000 mS
 > gm bound from Bandwidth Spec: 0.257 mS
 => SELECTED TARGET gm2 (MAX):   9.000 mS

--- Phase 4: Refining Sizing to Overcome Short-Channel Effects ---
Refinement complete. Achieved Baseline Gain: 0.902 V/V

--- Phase 5: Executing Power Scaling Loop ---
Scale  | ID (uA)   | W2 (um)   | W1 (um)   | Gain   | BW (MHz)  | Noise(uV)
-------------------------------------------------------------------------
1.0    | 1000.0    | 128.2     | 32.1      | 0.902  | 1175.8    | 33.25
2.0    | 500.0     | 256.5     | 64.1      | 0.878  | 778.2     | 27.03

[Constraint Broken] Loop Terminated:
 - Gain (0.878 V/V) < Target (0.900)
```

1. Define initial parameters: VDD = 1.8V, ID = 1mA, RL = 1000 ohms, CL = 2pF, unCox = 230uA/V^2, VTH = 0.366V.
2. Define constraint targets: Av >= 0.9, fp >= 100MHz, alpha = 0.5, L2 = 0.18um, L1 = 0.0.18um.
3. Calculate required input transconductance for gain: gm2 >= Av / (RL * (1 - Av)).
4. Calculate required input transconductance for bandwidth: gm2 >= (2 * pi * CL * fp) - (1 / RL).
5. Substitute bandwidth values: gm2 >= (2 * 3.14 * 2e-12 * 100e6) - 0.001 = 0.25 mS.
6. Find a third bound for gm2, using the input refered noise eq =8kt/3(1/gm1+gm1/gm2^2)
7. Select the dominant gm2 constraint to guarantee all specs: gm2 = max(9.0 mS, 0.25 mS,0.165 mS) = 9.0 mS.
8. Calculate required load transconductance using alpha: gm1 = alpha * gm2 = 0.5 * 9.0 mS = 4.5 mS.
9. Calculate input transistor width: W2 = L2 * (gm2^2 / (2 * unCox * ID)).
10. Substitute values for W2: W2 = 0.18um * (0.009^2 / (2 * 230e-6 * 1e-3)) = 31.7 um.
11. Calculate load transistor width: W1 = L1 * (gm1^2 / (2 * unCox * ID)).
12. Substitute values for W1: W1 = 0.54um * (0.0045^2 / (2 * 230e-6 * 1e-3)) = 23.7 um.
13. Set DC source voltage to exactly mid-rail for maximum symmetrical swing: VS = VDD / 2 = 0.9V.
14. Calculate input overdrive voltage: VOV2 = 2 * ID / gm2 = 2 * 1mA / 9.0mS = 0.222V.
15. Calculate required DC gate bias: VG = VS + VTH + VOV2 = 0.9V + 0.45V + 0.222V = 1.572V.
16. Calculate thermal spot noise density: vn^2 = (8 * k * T / (3 * gm2)) * (1 + alpha).
17. Substitute noise values at 300K: vn^2 = (8 * 4.14e-21 / (3 * 0.009)) * 1.5 = 1.84e-18 V^2/Hz.
18. Calculate total integrated RMS noise: Vrms = sqrt(vn^2 * fp).
19. Substitute RMS values: Vrms = sqrt(1.84e-18 * 100e6) = 13.5 uV.
20. Verify 13.5 uV is mathematically valid and safely below the 100 uV target specification.

The End